Two Ink Brackets

You are given a string s of even length containing equal numbers of opening brackets '[' and closing brackets ']'.

Each character in s is written in one of two inks, represented by a mask string m of the same length. If m[i] is 'P', the character at index i is written in permanent ink and cannot be modified or moved. If m[i] is 'p', the character is written in pencil.

You want to transform s into a balanced bracket sequence. A balanced bracket sequence is defined recursively as:

An empty string is balanced. If A is balanced, then [A] is balanced. If A and B are balanced, then AB is balanced.

You can perform the following operations on the pencil characters:

Swap any two pencil characters. This operation costs 1. Erase and rewrite a single pencil character in place, changing '[' to ']' or vice versa. This operation costs 2.

Note that rewrites may change the total bracket counts away from being equal; only the final string's balance matters.

Find the minimum total cost to make s balanced, or return −1 if it is impossible.

Input Format

The first line contains a string, s, denoting the bracket string containing '[' and ']'.

The second line contains a string, m, denoting the ink mask string containing 'P' and 'p'.

Constraints

2 <= |s| <= 10^5

2 <= |m| <= 10^5

Output Format

The output contains an integer value

Sample Input 0

```
[]
pp
```

Sample Output 0

```
0
```

Explanation 0

The input string is already a balanced bracket sequence, requiring zero operations and resulting in a minimum cost of 0.

Sample Input 1

```
][
PP
```

Sample Output 1

```
-1
```

Explanation 1

The string is unbalanced, and because both characters are written in permanent ink (P), they cannot be modified or swapped, making it impossible to balance.

Sample Input 2

```
][
pp
```

Sample Output 2

```
1
```

Explanation 2

Swapping the two pencil characters (p) transforms the unbalanced "][" into the balanced "[]" for a minimum cost of 1.